

## **Assignment - I** **Set-A**

**Course: B. Tech. Year/Semester: III & V      Session: 2025-2026**  
**Subject Name & Code: Design & Analysis of Algorithms (BCSC0059)**

**Date: 04 Sept.2025**

**Date of Submission: 25 Sept,2025**

1. Show that  $\frac{1}{2}n^2 + 3n = \Theta(n^2)$
2. What is the complexity of the following algorithm?  
    for ( $i = 0; i < n; i++$ )  
        for ( $j = 0; j < n; j++$ )  
             $a[i][j] = b[i][j] * x;$
3. Write an algorithm of Insertion Sort and compute the running time complexity. Also compute the complexity for increasing order sorted data, decreasing order sorted data and random data.
4. Suppose we want to search a number from a sorted list of numbers. Which searching algorithm is best suited for this task? Write a recurrence equation for this algorithm. Solve the recurrence equation.
5. Solve the recurrence  $T(n) = 2T(n/3) + n \lg n$
6. Can the master method be applied to the recurrence  $T(n) = 4T(n/2) + n^2 \lg n$ ? Why or why not?
7. Construct the binomial heap for the following sequence of numbers 7,2,4,17,1,11,6,8,15,10,20. Also apply the operation of extracting the minimum key from the binomial heap.
8. Insert the following elements in the order given into an empty B Tree of minimum degree 3  
  
    C N G A H E K Q M F W L T Z D P R X Y S  
  
    Undertake the following operations on the B Tree  
        a) Delete H      b) Delete T
9. Write the algorithm for counting sort. Illustrate the operation of counting sort on an array  $A = \{6,0,2,0,1,3,4,6,1,3,2\}$ .
10. Illustrate the procedure of BUILD-MAX-Heap on the array  
     $A = \{5, 3, 17, 10, 84, 19, 6, 22, 9\}$
11. Consider the Quick Sort algorithm in which the partitioning procedure splits elements into two sub-arrays and each sub-array contains at least one-fourth of the elements. Let  $T(n)$  be the number of comparisons required to sort an array of  $n$  elements. Then  $T(n) \leq ?$ .

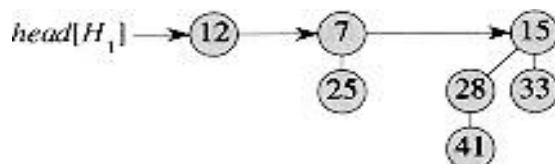
## Assignment - I Set-B

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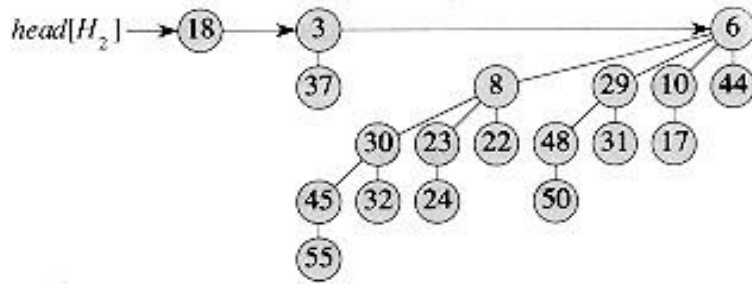
**Date: 04 Sept.2025**

**Date of Submission: 25 Sept,2025**

1. Solve the following recurrence:
  - a.  $T(n) = 2T(n/3) + n \lg n$
  - b.  $T(n) = 10T(n/3) + 17n^{1.2}$
  - c.  $T(n) = 16T(n/4) + n^3$ .
  - d.  $T(n) = T(n-1) + 1/n$  by iteration method.
2. To evaluate the performance of any algorithm, we compute time and space complexity. If an algorithm has recurrence  $T(n) = 2T(n/2) + n$  in best case and  $T(n) = T(n-q-1) + c$  in worst case, then compute the complexity in both the cases?
3. For parameters  $a$  and  $b$ , both of which are  $\omega(1)$ ,  
 $T(n) = T(n^{1/a}) + 1$ , and  $T(b) = 1$ .  
Then  $T(n)$  is
4. Write down the algorithm of a non-comparable sorting algorithm. Also sort the given data using it: 5, 3, 2, 4, 1.
5. Prove that any  $n$  key B-Tree of degree  $t \geq 2$ , have maximum height of  $\log_t (n+1)/2$ .
6. Construct the binomial heap for the following sequence of numbers 7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20. Also apply the operation of extracting the minimum key from the binomial heap.
7. Prove that  $f(n) = 100n + 5$  is not equal to  $\Omega(n^2)$ .
8. What do you understand by Divide-Conquer Approach? Write Quick Sort algorithm and apply it to the given data 12, 32, 9, 5, 2, 4. Also analyze the complexity if data is already sorted.
9. Discuss about the Binomial Heap and its properties. Create a binomial heap using given (a) and (b) binomial heaps.

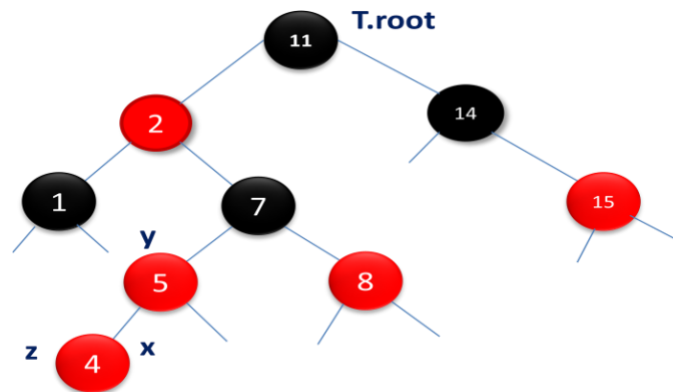


**(a)**



(b)

10. An element  $z$  has been inserted into a Red-Black Tree in which 2, 4, 5, 8 and 15 are red nodes and others are black. After insertion, the rules of Red-Black Tree is violated, perform the suitable operation to satisfy all the properties of Red-Black Tree.



11. Insert the following numbers in B-Tree of minimum degree 2.

34,22,32,11,21,14,16,2,4,7,1,18,44,33,65,78,76,29,50,40,30,45

Then, delete given keys in order:

- a) 11   b) 22   c) 44   d) 65   e) 33

## Assignment - I

### Set-C

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**Date of Submission: 25 Sept,2025**

- 1.** Consider the following three functions.  
 $f_1 = 10^n$ ,  $f_2 = n^{\log n}$ ,  $f_3 = n^{\sqrt{n}}$   
Which one of the following options arranges the functions in the increasing order of asymptotic growth rate?  
**a.**  $f_3, f_2, f_1$                                       **b.**  $f_2, f_1, f_3$   
**c.**  $f_1, f_2, f_3$                                       **d.**  $f_2, f_3, f_1$
- 2.** Solve the following recurrence:  
**a.**  $T(n) = 4T(n/2) + n2\sqrt{n}$   
**b.**  $T(n) = T(n-2) + 2 \lg n$   
**c.**  $T(n) = T(n-1) + \lg n$ .
- 3.** There is a huge amount of data of a university; a sorting algorithm is required to apply over the data to sort according to the student's roll number. Which sorting algorithm you will apply to arrange the data in ascending order? Also derive the complexity in best, worst and average case complexity.
- 4.** Compute the complexity of given recurrence:  $T(n) = 2T(\sqrt{n}) + 1$
- 5.** Show that for any real constant  $a$  and  $b$ , where  $b > 0$ ,  
 $(n+a)^b = \theta(n^b)$
- 6.** Find  $\Theta$  bound for  $f(n) = n^2/2 - n/2$ .
- 7.** Apply Radix sort on  $A = \{CAT, BAT, COW, SIT, MAY, WHY\}$ .
- 8.** Write algorithm of Heap sort and apply Max\_Heap\_Build and Heapify to sort the given data: 3, 2, 5, 1, 9, 7, 11, 4, 20, 6. Also, compute the worst case complexity.
- 9.** Construct a red-black tree by inserting the keys in the following sequence into an initially empty red-black tree: 13, 10, 8, 3, 4 and 9.
- 10.** Write down the properties of Red-Black Tree with a suitable example that satisfy all the properties.
- 11.** Discuss the stable property. Perform the sorting operation and show the stable property on given data:  $A = \{5, 0, 5, 0, 1, 7, 4, 7, 1, 3, 2\}$ .
- 12.** What are the various applications of B-Trees? Create a B-Tree for the following data- 23, 2, 50, 34, 45, 100, 88, 67, 77, 47, 37, 105, 55, 84, and 99 where degree  $t=3$ .

## Assignment - I Set-D

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**Date: 04 Sept.2025**

**Date of Submission: 25 Sept,2025**

1. What will be the recurrence relation of the following code?

```
int sum(int n)
{
    If(n==1)
        return 1;
    else
        return n+sum(n-1);
}
```

2. Compute the time complexity of the given algorithm and also find the complexity for given data 1, 2, 3, 4, 5, 6, 7, 8, 9 for the same algorithm.

**Algorithm(arr[], n) //arr is array, n is size of array**

```
{
    for z=1 to n do
    {
        x = arr[z];
        y = z - 1;
        while (y >= 0 && arr[j] > x) do {
            arr[j + 1] = arr[j];
            y = y - 1;
        }
        arr[y + 1] = x;
    }
}
```

3. Show the red-black trees that result after successively inserting the keys 41,38,31,12,19,8 into an initially empty red-black tree.
4. Show that quicksort's best-case running time is  $\Omega(n \log)$ .
5. Is the array with values <23,17,14,6,13,10,1,5,7,12> a max heap?
6. Draw the B-tree of degree 3 created by inserting data arriving in sequence from the two sets shown below. Compare the two B-trees to determine whether the order of data will create different B-tree.

Data 1: 89 78 8 19 20 33 56 44

Data 2: 44 56 33 20 19 8 78 89

7. Do you think Quick sort is stable or not? Why or why not?
8. Is the Quick sort suitable for use on large datasets? Why or why not?

9. Write an algorithm of Shell sort and apply it to sort the given data: {5, 3, 15, 10, 19, 7, 11, 4, 25, and 61}
10. Banks often record transactions on an account in order of the times of the transactions, but many people like to receive their bank statements with checks listed in order by check

number. People usually write checks in order by check number, and merchants usually cash them with reasonable dispatch. The problem of converting time-of-transaction ordering to check-number ordering is therefore the problem of sorting almost sorted input. Argue that the procedure Insertion-sort would tend to beat the procedure QuickSort on this problem.



## Assignment - I Set-E

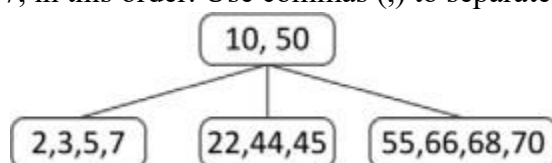
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1. What will be the recurrence relation of the following code?  

```
int xpowy(int x, int n)
if (n==0) return 1;
if (n==1) return x;
if ((n % 2) == 0)
return xpowy(x*x, n/2);
else
return xpowy(x*x, n/2) * x;
```
2. Give an expression for the runtime complexity  $T(n)$  if the recurrence can be solved with the Iterative Method
  1.  $T(n) = T(n-1) + \log n$
  2.  $T(n) = 2T(n/2) + n$
3. Give an expression for the runtime complexity  $T(n)$  if the recurrence can be solved with the Master Theorem
  - $T(n) = 2T(n/4) + \sqrt{n}$
  - $T(n) = 3T(n/5) + n$
  - $T(n) = T(n/2) + 1$
4. Generate a recursion tree for the recurrence  $T(n) = T(n/3) + T(2n/3) + n$
5. Create B tree of degree 3 for the following sequence of keys.  
40, 21, 30, 56, 17, 19, 48, 29, 24
6. Draw the B-tree of order 3 created by inserting data arriving in sequence from the two sets shown below. Compare the two B-trees to determine whether the order of data will create different B-tree.  
Data 1: 89 78 8 19 20 33 56 44  
Data 2: 44 56 33 20 19 8 78 89
7. Given the following B-tree of order  $m = 5$ , show each corresponding B-tree after insertion of 17, 6, 21, 67, in this order. Use commas (,) to separate the data in a node.



- 8.** Construct a red-black tree by inserting the keys in the following sequence into an initially empty red-black tree: 13, 10, 8, 3, 4 and 9.
- 9.** Analyze that the Worst Case time complexity of the quick sort algorithm is  $O(n^2)$
- 10.** Find out the time complexity of the following code:  
`int i, j, k = 0;`

```
for (i = n / 2; i <= n; i++) {  
    for (j = 2; j <= n; j = j * 2) {  
        k = k + n / 2;  
    }  
}
```